

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا،

وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا،

Simple Linear Regression

Week 13

Variables

i. Tar content and inlet temperature

ii. Gas mileage and engine volume

iii. Price of house and square feet of living space

iv. Marks of students and no of study hours

v. Runoff of the river and precipitation

vi. Blow counts and compressive strength of stiff clay

Dependent/ Response

Independent/
Regressors

Relationship

Dependent/ Response (Y)

Independent/ Regressor (X)

$$Y = \beta_0 + \beta_1 x,$$

Deterministic

$$Y = \beta_0 + \beta_1 x + \epsilon.$$

Probabilistic

Regression Analysis

- i. Simple regression
- ii. Multiple regression

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

Hydrogen loss (Y)

Storage time (X1)

Storage
temperature (X2)

The Simple Linear regression (SLR) Model

- Single regressor with relationship between y and x is linear.
- Random component that relates to the variables.

$$(x_1, y_1), \dots, (x_n, y_n)$$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

$$N(0, \sigma^2)$$

$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

Parameters

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Intercept

Slope
parameter

$$E(Y_i) = \beta_0 + \beta_1 x_i$$

Simple Linear Regression Model

The **simple linear regression** model

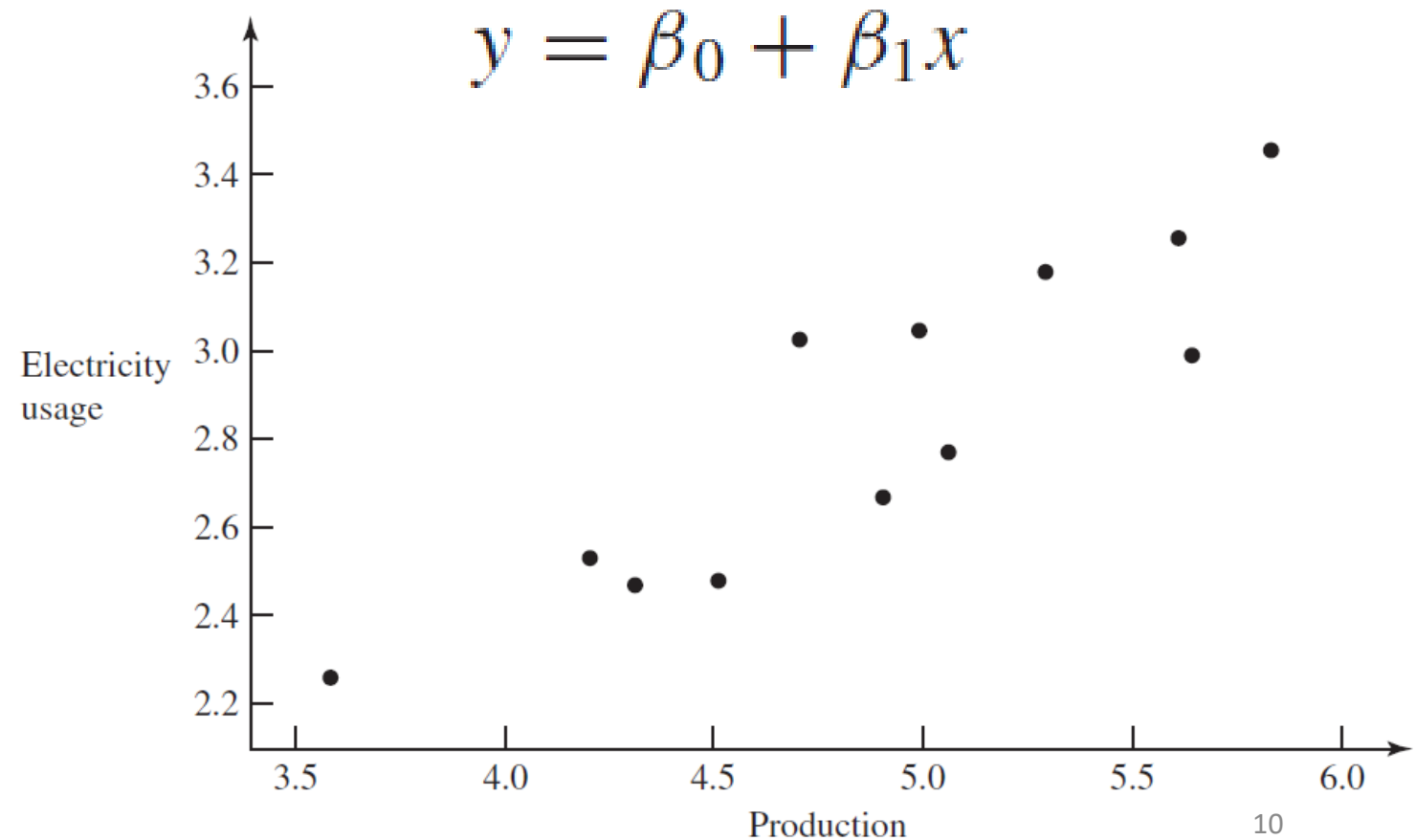
$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

fits a straight line through a set of paired data observations $(x_1, y_1), \dots, (x_n, y_n)$. The error terms $\epsilon_1, \dots, \epsilon_n$ are taken to be independent observations from a $N(0, \sigma^2)$ distribution. The three unknown parameters, the **intercept parameter** β_0 , the **slope parameter** β_1 , and the **error variance** σ^2 , are estimated from the data set.

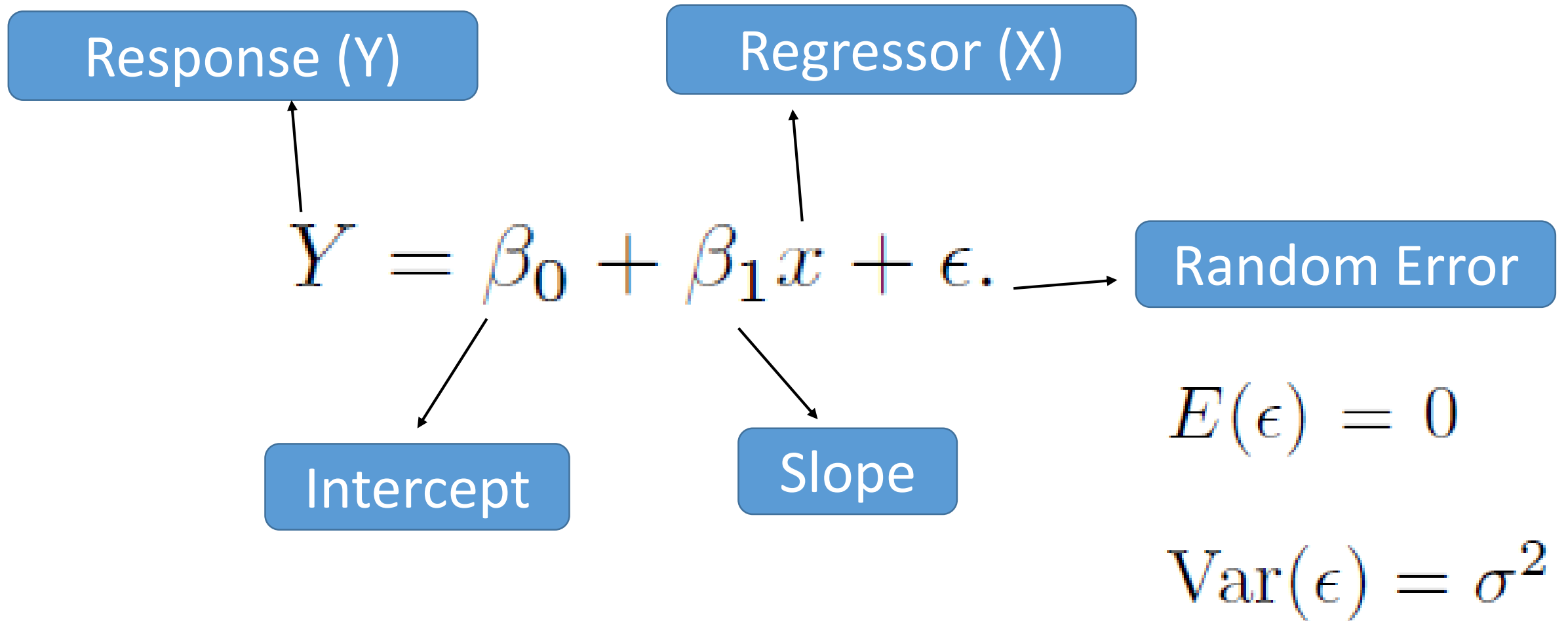
Example

The manager of a car plant wishes to investigate how the plant's electricity usage depends upon the plant's production. The data set shown in Figure 12.5 is compiled and provides the plant's production and electrical usage for each month of the previous year. The electrical usage is in units of a million kilowatt-hours, and the production is measured as the value in million-dollar units of the cars produced in that month.

	Production (\$ million)	Electricity usage (million kWh)
January	4.51	2.48
February	3.58	2.26
March	4.31	2.47
April	5.06	2.77
May	5.64	2.99
June	4.99	3.05
July	5.29	3.18
August	5.83	3.46
September	4.70	3.03
October	5.61	3.26
November	4.90	2.67
December	4.20	2.53



Simple Linear Regression Model (SLRM)



Important Aspects about SLRM

1. Y is random
2. X is fixed
3. Homogeneous variance
4. Probabilistic model
5. Y -values distributed around true regression
6. Estimated parameters

The fitted Regression Line

$$Y = \beta_0 + \beta_1 x + \epsilon.$$

SLRM

$$\hat{y} = b_0 + b_1 x,$$

Fitted regression line

Predicted/ Fitted value



Estimates

Method of Least Squares

We shall find b_0 and b_1 , the estimates of β_0 and β_1 , so that the sum of the squares of the residuals is a minimum. The residual sum of squares is often called the sum of squares of the errors about the regression line and is denoted by SSE . This

minimization procedure for estimating the parameters is called the **method of least squares**. Hence, we shall find a and b so as to minimize

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

Differentiating SSE with respect to b_0 and b_1 , we have

$$\frac{\partial(SSE)}{\partial b_0} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i), \quad \frac{\partial(SSE)}{\partial b_1} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) x_i.$$

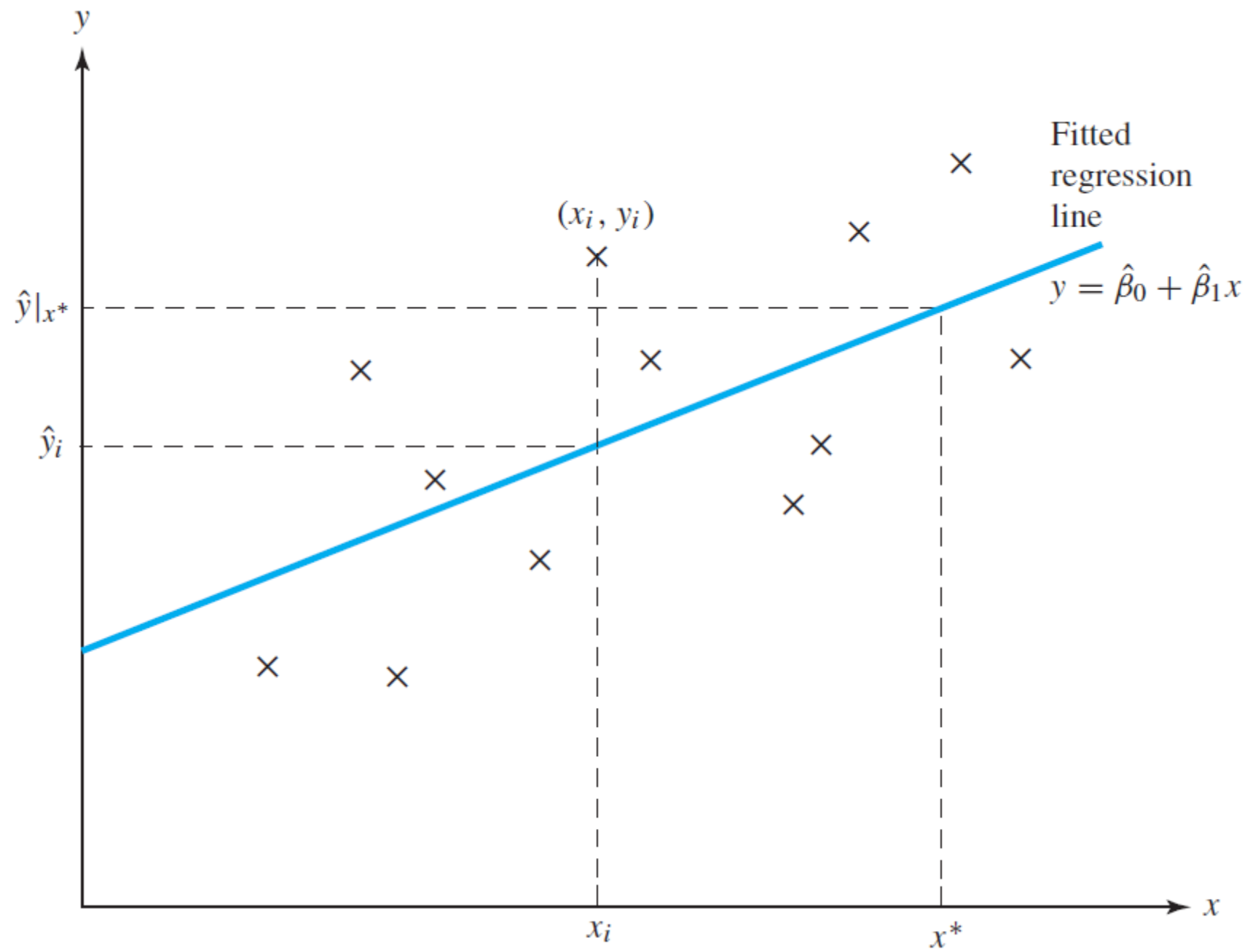
Setting the partial derivatives equal to zero and rearranging the terms, we obtain the equations (called the **normal equations**)

$$nb_0 + b_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i, \quad b_0 \sum_{i=1}^n x_i + b_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

$$nb_0 + b_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i, \quad b_0 \sum_{i=1}^n x_i + b_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

$$b_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \text{ and}$$

$$b_0 = \frac{\sum_{i=1}^n y_i - b_1 \sum_{i=1}^n x_i}{n} = \bar{y} - b_1 \bar{x}.$$



Estimate the regression line for the pollution data

Table 11.1: Measures of Reduction in Solids and Oxygen

Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)	Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)
3	5	36	34
7	11	37	36
11	21	38	38
15	16	39	37
18	16	39	36
27	28	39	45
29	27	40	39
30	25	41	41
30	35	42	40
31	30	42	44
31	40	43	37
32	32	44	44
33	34	45	46
33	32	46	46
34	34	47	49
36	37	50	51
36	38		

x	y	xx	xy
3	5	9	15
7	11	49	77
11	21	121	231
15	16	225	240
18	16	324	288
54	69	728	851

$$\sum_{i=1}^{33} x_i = 1104, \quad \sum_{i=1}^{33} y_i = 1124, \quad \sum_{i=1}^{33} x_i y_i = 41,355, \quad \sum_{i=1}^{33} x_i^2 = 41,086$$

$$b_1 = \frac{(33)(41,355) - (1104)(1124)}{(33)(41,086) - (1104)^2} = 0.903643 \text{ and}$$

$$b_0 = \frac{1124 - (0.903643)(1104)}{33} = 3.829633.$$

$$b_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}.$$

Thus, the estimated regression line is given by

$$\hat{y} = 3.8296 + 0.9036x.$$

Task: 1

A mathematics placement test is given to all entering freshmen at a small college. A student who receives a grade below 35 is denied admission to the regular mathematics course and placed in a remedial class. The placement test scores and the final grades for 20 students who took the regular course were recorded.

- i. Plot a scatter diagram.
- ii. Find the equation of the regression line to predict course grades from placement test scores.
- iii. Graph the line on the scatter diagram.

Placement Test	Course Grade
50	53
35	41
35	61
40	56
55	68
65	36
35	11
60	70
90	79
35	59
90	54
80	91
60	48
60	71
60	71
40	47
55	53
50	68
65	57
50	79

Task: 2

A professor in the School of Business in a university polled a dozen colleagues about the number of professional meetings they attended in the past five years (x) and the number of papers they submitted to refereed journals (y) during the same period. The summary data are given as follows:

$$n = 12, \quad \bar{x} = 4, \quad \bar{y} = 12,$$
$$\sum_{i=1}^n x_i^2 = 232, \quad \sum_{i=1}^n x_i y_i = 318.$$

Fit a simple linear regression model between x and y by finding out the estimates of intercept and slope. Comment on whether attending more professional meetings would result in publishing more papers.